



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-044A MetroParcel Orchestration Gate

Protected independent phase Gate A.

Artifact	C3-044A
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.

MetroParcel scenario

MetroParcel receives daily parcel-event partitions. A late historical partition contains an older version of an entity that was updated later. Build an operator-safe Airflow workflow and prove retries, reprocessing and partial recovery cannot duplicate or regress current state.

Begin only after RapidRoute Mini-Lab PASS and all lesson mastery evidence is complete.

Independent rules

- Use supplied case ZIP; do not edit fixtures to make controls pass.
- External lesson/review/video answers are closed during attempt.
- State logical interval and side-effect contract before implementation.
- Save actual DAG/code/config plus run/task/interval and independent reconciliation evidence.
- No secret/credential value belongs in code or evidence.

Required deliverables

- Fresh DAG with explicit wait -> extract -> validate -> load/merge -> reconcile -> publish boundaries.
- Correct daily logical interval + timezone-aware start; prove historical rerun path stability.
- Separate Params/config/Connection IDs/secrets; sanitize evidence.
- One bounded transient retry + one deterministic failure that stays failed.
- Bounded readiness wait + missing/late input recovery.
- Named pool for case sink + deliberately bounded backfill concurrency.
- Process/reprocess all supplied dates; prove stable version semantics and same-date idempotence.
- Layered predeploy tests/static checks + real learner-Airflow discovery/run evidence.
- Diagnose one injected incident from exact run/task/interval evidence; perform smallest safe recovery.
- Concise operator runbook + post-recovery reconciliation.

Competency Gate - every required competency must pass

Competency	PASS evidence
DAG architecture and meaningful task boundaries	Observed, validated, reproducible and explainable on this fresh case.
Interval correctness and historical partition mapping	Observed, validated, reproducible and explainable on this fresh case.
Failure classification, bounded retry/timeout	Observed, validated, reproducible and explainable on this fresh case.
Idempotent same-date rerun and stale-safe historical reprocessing	Observed, validated, reproducible and explainable on this fresh case.
Readiness wait and scarce-resource controls	Observed, validated, reproducible and explainable on this fresh case.
Testing/discovery/troubleshooting evidence	Observed, validated, reproducible and explainable on this fresh case.
Operator runbook and professional handoff	Observed, validated, reproducible and explainable on this fresh case.

PASS only if every required competency passes AND zero critical failures remain. Every required competency stands on its own; one failure blocks PASS.

Evidence packet

Evidence	Your saved proof/path/note
DAG/task graph	
Schedule + interval mapping	
Sanitized config/connection/pool	
Normal run evidence	
Transient retry evidence	
Missing/late input behavior	
Historical backfill/reprocess evidence	
Same-date/partial rerun idempotence	
Pre-deploy test evidence	
Reconciliation controls	
Incident troubleshooting note	
Runbook/handoff	